

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import statsmodels.api as sm

print("All good!")
```

All good!

```
df = pd.read_csv('marketing_and_sales_data_evaluate_lr.csv')
print(df.shape) # How many rows and columns?
print(df.head()) # First 5 rows
print(df.isnull().sum()) # Any missing values?
```

```
(4572, 4)
   TV      Radio  Social_Media    Sales
0  16.0   6.566231    2.907983  54.732757
1  13.0   9.237765    2.409567  46.677897
2  41.0  15.886446    2.913410  150.177829
3  83.0  30.020028    6.922304  298.246340
4  15.0   8.437408    1.405998   56.594181
TV      10
Radio      4
Social_Media  6
Sales      6
dtype: int64
```

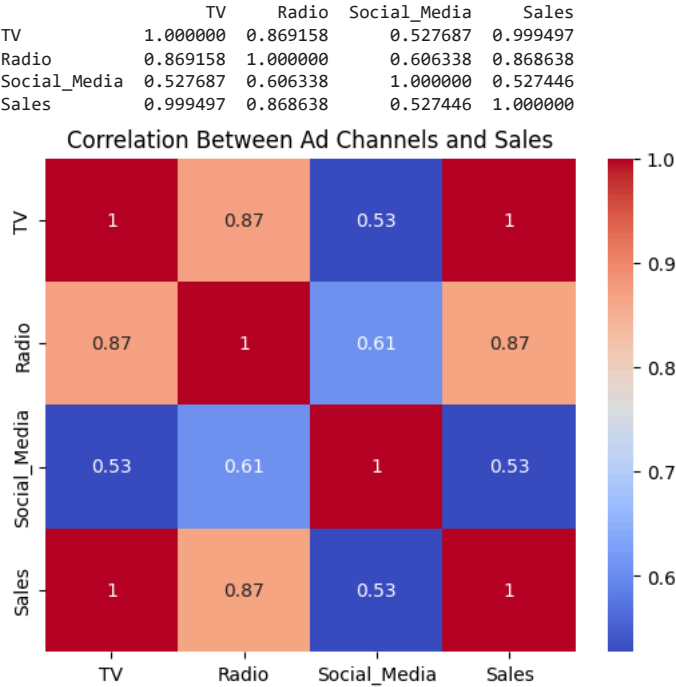
```
# Remove rows with missing values
df = df.dropna()
```

```
# Confirm they're gone
print(df.isnull().sum())
print(df.shape)
```

```
TV      0
Radio    0
Social_Media  0
Sales    0
dtype: int64
(4546, 4)
```

```
# Check correlation of everything with Sales
print(df.corr())
```

```
# Visual heatmap
sns.heatmap(df.corr(), annot=True, cmap='coolwarm')
plt.title('Correlation Between Ad Channels and Sales')
plt.show()
```



```
# X = TV (our predictor), y = Sales (what we're predicting)
X = df['TV']
y = df['Sales']

# Add intercept
X = sm.add_constant(X)

# Build and fit the model
model = sm.OLS(y, X).fit()

# Show results
print(model.summary())
```

OLS Regression Results						
=====						
Dep. Variable:	Sales	R-squared:	0.999			
Model:	OLS	Adj. R-squared:	0.999			
Method:	Least Squares	F-statistic:	4.517e+06			
Date:	Thu, 11 Jun 2026	Prob (F-statistic):	0.00			
Time:	16:26:49	Log-Likelihood:	-11366.			
No. Observations:	4546	AIC:	2.274e+04			
Df Residuals:	4544	BIC:	2.275e+04			
Df Model:	1					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	-0.1325	0.101	-1.317	0.188	-0.330	0.065
TV	3.5615	0.002	2125.272	0.000	3.558	3.565
=====						
Omnibus:	0.052	Durbin-Watson:	1.998			
Prob(Omnibus):	0.974	Jarque-Bera (JB):	0.031			
Skew:	-0.001	Prob(JB):	0.985			
Kurtosis:	3.012	Cond. No.	138.			
=====						

Notes:

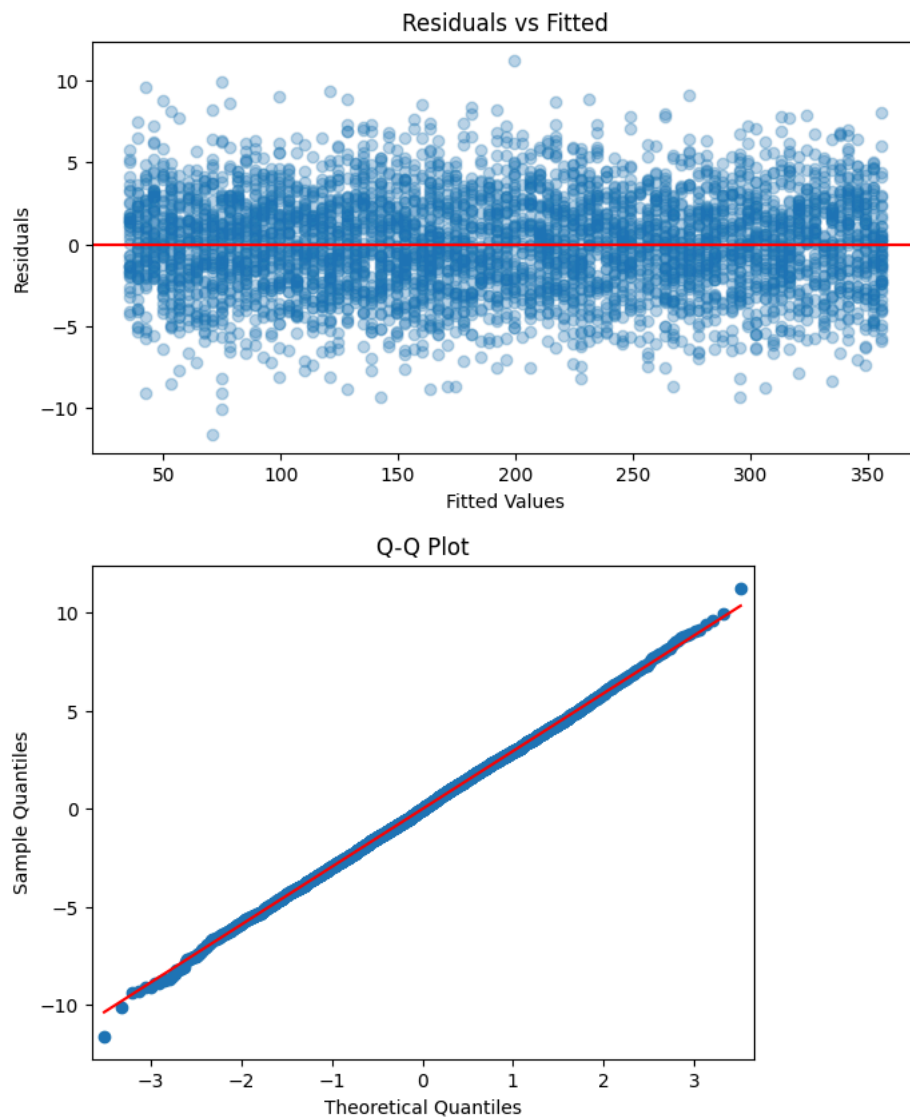
[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

```
fitted = model.fittedvalues
residuals = model.resid

# Plot 1: Residuals vs Fitted
plt.figure(figsize=(8,4))
plt.scatter(fitted, residuals, alpha=0.3)
plt.axhline(0, color='red')
plt.title('Residuals vs Fitted')
plt.xlabel('Fitted Values')
plt.ylabel('Residuals')
```

```
plt.show()

# Plot 2: Q-Q Plot (Normality check)
sm.qqplot(residuals, line='s')
plt.title('Q-Q Plot')
plt.show()
```



```
# Histogram of residuals
plt.figure(figsize=(8,4))
plt.hist(model.resid, bins=30, color='steelblue', edgecolor='black')
plt.title('Distribution of Residuals')
plt.xlabel('Residuals')
plt.ylabel('Frequency')
plt.show()
```

